

CCNA 200-301 V2.0

04 IPv4 Addressing and Subnetting

Part I — Foundations

EXAM TOPICS

1.3

LECTURER

Dr. Mustafa Hameed

THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Information Technology

What you will be able to do

- **Convert** between dotted decimal, binary and prefix length fast enough to do it in an exam without a calculator.
- **Compute** the network address, broadcast address, usable range and host count of any subnet, in under thirty seconds.
- **Subnet** a given block to meet a stated requirement, and apply variable-length masks without overlapping.
- **Diagnose** the four addressing faults that account for almost every broken host: wrong mask, wrong gateway, address outside the subnet, and duplicate address.

Before we start

Chapter 1 and Chapter 3. You need to be comfortable that a switch works on MAC addresses and a router works on IP addresses.

Vocabulary for this chapter

dotted decimal

prefix length

subnet mask

network address

broadcast address

usable range

block size

magic number

RFC 1918

private address

public address

VLSM

default gateway

APIPA

Each is defined where it first matters — not here.

This topic is worth more time than any other in Part I

This is what the blueprint actually asks for

Topic **1.3** says *troubleshoot* IPv4 addressing, assignment and subnetting, “public and private”. You will be given a host that cannot reach something and expected to spot that its mask is wrong. Every second you spend computing a subnet boundary is a second not spent reasoning, so the arithmetic has to be automatic. Learn the method in §sec:magic until it is boring.

Bits, bytes and octets

A **bit** is a single 0 or 1. Eight bits make a **byte**, which in networking is almost always called an **octet** — the words mean the same thing here.

Each position in an octet is worth twice the one to its right:

Both directions, in one line each

Binary to decimal: add the position values wherever there is a 1. $64+8+4+1 = 77$.

Decimal to binary: walk left to right, and at each position ask “does this fit in what is left?”. For 77: 128 no (0), 64 yes (1, leaving 13), 32 no (0), 16 no (0), 8 yes (1, leaving 5), 4 yes (1, leaving 1), 2 no (0), 1 yes (1, leaving 0). That gives **01001101**.

Two facts fall straight out and are worth holding on to. **255 is all eight bits set** — which is why it is the largest value an octet can hold, and why a mask octet of 255 means “all of this octet is network”. And **0 is all eight clear**, meaning “none of it is”.

Where the block size comes from

Take 192.168.10.77 with mask 255.255.255.192, and write only the last octet of each in bits:

{@}||@} address 77 & 01 001101 mask 192 & 11 000000

The mask has **two** 1 bits in this octet, so two bits of it are network and the remaining **six** are host. Set those six host bits to zero and you have the network address; set them all to one and you have the broadcast address:

{@}|||@} network & 01 000000 & = 64 broadcast & 01 111111 & = 127

So the subnet is 192.168.10.64 to 192.168.10.127 — and it is **64 addresses** wide, because six free host bits give $2^6 = 64$ combinations.

Now notice: $256 - 192 = 64$ as well. That is not a coincidence and it is not a trick. Subtracting the mask value from 256 is simply a fast way of asking “how many combinations do the remaining host bits have?” **That is the magic number**, and Section sec:magic does the whole job with it.

Powers of two, which you should know on sight

Host bits decide subnet size, so these eleven numbers cover almost every question you will be asked.

{@}|||||||@} 2^1 & 2^2 & 2^3 & 2^4 & 2^5 & 2^6 & 2^7 & 2^8 & 2^9 & 2^{10} & 2^{11} 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & 2048

Usable hosts is always two fewer than the total, because the all-zeros address names the network itself and the all-ones address is the broadcast. Neither can be given to a host. So six host bits give 64 addresses and **62** usable ones.

Network part and host part

The **subnet mask** marks, with 1 bits, which leading bits of the address identify the **network**. The remaining 0 bits identify the **host** within it. Written as a count of 1 bits, the mask is a **prefix length**: 255.255.255.0 and /24 are the same statement.

Two addresses are on the same subnet if and only if their network parts are identical. That single sentence is the source of most addressing faults: a host with the wrong mask computes a different network part, decides its neighbour is remote, and sends the traffic to a gateway that may not want it.

Default gateway

The address a host sends traffic to when the destination is **not** on its own subnet. Almost always a router or a layer 3 switch interface on that subnet.

Every host makes exactly one decision for every packet it sends, and it makes it with the mask:

Three consequences that account for a great many faults, and that Chapter 28 will make you test for directly:

- **Same network part as mine?** Then the destination is local. ARP for it directly and send the frame straight there. The gateway is not involved at all.
- **Different?** Then it is remote. ARP for the *gateway*, and hand the frame to it — with the destination's IP address still inside, but the gateway's MAC address on the outside.
- **A host with no gateway can reach its own subnet and nothing else.** Everything local works, so the user reports “the internet is down” rather than “my machine is misconfigured”.
- **The gateway must be inside the host's own subnet.** A host will not ARP for something it believes is remote, so a gateway outside the subnet is unreachable by definition — which is the fault in this chapter's Break & Fix.

Default gateway (continued)

- **Get the mask wrong and the host makes the local-or-remote decision wrongly**, for some destinations and not others. That is why a wrong mask presents as “some things work”, which is far harder to read than “nothing works”.

Private addresses — RFC 1918

Three ranges are reserved for internal use and are never routed on the internet. The exam expects you to recognise them instantly.

- **10.0.0.0/8** — one very large block
- **172.16.0.0/12** — that is **172.16.0.0** to **172.31.255.255**, and the upper bound catches people out
- **192.168.0.0/16**

Also worth knowing: **169.254.0.0/16** is **APIPA**, which a host assigns itself when DHCP fails. A host with a 169.254 address has not been given one — see Chapter 20.

Four steps, every time

- **Find the interesting octet** — the one where the mask is neither 255 nor 0.
- **Block size** = $256 - (\text{the mask value in that octet})$. This is the magic number.
- **Count in blocks** from zero in that octet until you pass the address. The multiple you last passed is the **network address**.
- **Broadcast** = next network address - 1. Usable range is everything between the two, exclusive.

Where does 172.16.83.190/26 live?

- Mask for /26 is 255.255.255.192. The interesting octet is the fourth.
- Block size = $256 - 192 = 64$.
- Count in 64s in the fourth octet: 0, 64, 128, **192**. Our host is 190, which sits after 128 and before 192. Network address is 172.16.83.128.
- Next network is 172.16.83.192, so broadcast is 172.16.83.191. Usable range is 172.16.83.129 to 172.16.83.190.

So .190 is the *last usable address* in its subnet — a perfectly valid host address, and a favourite exam distractor because it looks like a broadcast.

The same method in the third octet: 10.44.150.7/21

- /21 is 255.255.248.0. Interesting octet is the third.
- Block size = $256 - 248 = 8$.
- Count in 8s in the third octet: 0, 8, ..., 144, **152**. Our 150 sits after 144. Network is 10.44.144.0.
- Next network is 10.44.152.0, so broadcast is 10.44.151.255. Usable is 10.44.144.1 to 10.44.151.254 — 2046 hosts.

Note that the fourth octet plays no part in finding the boundary. Only the interesting octet does.

Split 192.168.20.0/24 for four sites of 50, 25, 10 and 2 hosts

Work *largest first*, or you will strand addresses.

- 50 hosts needs 62 usable → /26, block 64. 192.168.20.0/26 — usable .1–.62.
- 25 hosts needs 30 usable → /27, block 32. Next free address is .64: 192.168.20.64/27 — usable .65–.94.
- 10 hosts needs 14 usable → /28, block 16. Next free is .96: 192.168.20.96/28 — usable .97–.110.
- 2 hosts (a router link) → /30, block 4. Next free is .112: 192.168.20.112/30 — usable .113–.114.

Everything from .116 upwards is still free. This is **VLSM**: different masks inside one block, sized to the requirement.

The two rules VLSM must obey

- **A subnet must start on a multiple of its own block size.** A /27 can start at .64 or .96, never at .70.
- **Subnets must not overlap.** If you allocate smallest-first you will eventually place a large subnet where it does not fit, and have to start again. Largest first avoids this.

An interface address and a default gateway

Configure

```
! On a router interface -- the mask is written in full.
interface GigabitEthernet0/0
  description User VLAN 10 gateway
  ip address 192.168.20.1 255.255.255.192
  no shutdown
!
! On a Layer 2 switch, the management address lives on an SVI,
! and the switch needs a default gateway because it does not route.
interface Vlan10
  ip address 192.168.20.2 255.255.255.192
  no shutdown
!
ip default-gateway 192.168.20.1
```

show ip interface brief

```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.20.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	down	down

Reading the two status columns

The pair of columns is the single most useful diagnostic on a router, and the combinations mean different things.

- **up / up** — working.
- **up / down** — the cable and electrical link are fine, but the protocol is not. Usually an encapsulation or keepalive mismatch, or the far end is not configured.
- **down / down** — a physical problem. Cable, transceiver, or the other end is shut. Go back to Chapter 2.
- **administratively down / down** — somebody typed `shutdown`. The fix is `no shutdown`.

Configuring and verifying an address

- **Forgetting no shutdown.** Router interfaces are shut by default. Switch ports are not. This asymmetry catches everyone once.
- **Assuming 172.16.0.0/12 stops at 172.16.** It runs to 172.31.255.255. 172.32.0.0 is public.
- **Giving the gateway an address outside the subnet.** A host will accept the configuration and then fail silently for everything off-net.
- **Using the network or broadcast address on a host.** .0 and .255 are only network and broadcast in a /24. In a /26 the boundaries move, and 192.168.20.64 is a network address even though it does not end in zero.

“I can reach my own floor but nothing else”.

A host is configured `192.168.20.100` mask `255.255.255.192`, gateway `192.168.20.1`. It can ping `192.168.20.90` but not its gateway and nothing beyond.

“I can reach my own floor but nothing else”.

What would you check first — and what do you expect to see?

Why it is doing that

Diagnosis. Work the mask. /26 gives block size 64, so the boundaries are .0, .64, .128, .192. The host at .100 is in the 192.168.20.64/26 subnet, usable .65–.126. Its gateway at .1 is in the *first* subnet, 192.168.20.0/26. The host and its gateway are on different subnets, so the host cannot ARP for it and has no way off its own segment. It can still reach .90 because that address is in the same /26.

And the lesson

Fix. Either move the host into the subnet its gateway serves, or — far more likely what was intended — correct the gateway to **192.168.20.65**, the first usable address of the subnet the host is actually in. Then confirm with a ping to the gateway before testing anything remote.

Subnet, configure, then break it

Topology. One router, two switches, four PCs — two per VLAN.

1. Subnet `172.16.5.0/24` into two subnets of at least 100 hosts each. Write down, before touching a device: network address, first and last usable address, and broadcast address for both.
2. Configure the router with a gateway address in each subnet and the PCs with addresses from your plan. Verify with `show ip interface brief` and a ping within each subnet.
3. Prove routing works between the two subnets by pinging across.
4. Now change one PC's mask from your planned value to `255.255.255.0`. Predict what will break *before* you test, then test it. Explain the result in terms of the network part.

Subnet, configure, then break it (continued)

1. Change it back, then set one PC's gateway to an address in the other subnet. Predict and test again.
2. **Extension.** Re-plan the same /24 using VLSM for four segments of 100, 50, 20 and 2 hosts, and configure it without overlaps.

CHECKPOINT 1 of 3

For 10.19.200.37/27: give the network address, broadcast address, and first and last usable address.

Discuss · then advance

Checkpoint 1 of 3

For 10.19.200.37/27: give the network address, broadcast address, and first and last usable address.

/27 gives a block size of 32 in the fourth octet. The address ends in **37**, which falls in the block starting at 32. Network **10.19.200.32**, broadcast **10.19.200.63**, first usable **10.19.200.33**, last usable **10.19.200.62**.

CHECKPOINT 2 of 3

A host is 192.168.1.130/25 and its gateway is 192.168.1.126.
Will it work? Show your reasoning.

Discuss · then advance

Checkpoint 2 of 3

A host is 192.168.1.130/25 and its gateway is 192.168.1.126. Will it work? Show your reasoning.

No. /25 has a block size of 128, so the host is in 192.168.1.128/25, whose range is .129 to .254. The gateway 192.168.1.126 is in the *other* half, 192.168.1.0/25. The host will ARP for a gateway that is not on its segment and reach nothing beyond its own subnet.

CHECKPOINT 3 of 3

How many usable addresses does a /30 provide, and why is a /31 sometimes used instead on a router-to-router link?

Discuss · then advance

Checkpoint 3 of 3

How many usable addresses does a /30 provide, and why is a /31 sometimes used instead on a router-to-router link?

Two. A /31 is used on a router-to-router link because there are only ever two devices and neither needs a broadcast address, so it provides both addresses usefully and halves the address consumption of every point-to-point link on a network.

What to take away

- Address plus mask equals two facts: which network, and which host. Devices are on the same subnet only if their network parts match.
- Block size = 256 - mask octet. Count in blocks. That method answers every subnetting question you will be asked.
- Usable = block - 2, except /31 and /32.
- Private ranges are 10/8, 172.16/12 (to 172.31) and 192.168/16. A 169.254 address means DHCP failed.
- Plan VLSM largest first, and start every subnet on a multiple of its own block size.

What to take away (continued)

- Four faults cover most cases: wrong mask, wrong gateway, address outside the subnet, duplicate address.

Where we got to

- Address plus mask equals two facts: which network, and which host. Devices are on the same subnet only if their network parts match.
- Block size = $256 - \text{mask octet}$. Count in blocks. That method answers every subnetting question you will be asked.
- Usable = block - 2, except /31 and /32.
- Private ranges are 10/8, 172.16/12 (to 172.31) and 192.168/16. A 169.254 address means DHCP failed.

NEXT

Chapter 5 — IPv6 Addressing